Analysis of INC 5000 Companies and results using SQL

Please, note that queries with results more than 10 rows are set to limit of 10 rows on purpose.

-- DATA PRE_PROCESSING

-- 1. CALCULATING TOTAL NUMBER OF UNIQUE COMPANIES

```
SELECT (COUNT(DISTINCT name)) AS name_count
FROM inc_5000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name_count
4988

-- EXPLORATORY DATA ANALYSIS
-- 1. SHOWING WHICH INDUSTRIES ARE MOST AND LEAST REPRESENTED IN THE LIST

```
SELECT industry,
(COUNT(*)) AS industry_count
FROM inc_5000
GROUP BY industry
ORDER BY industry_count DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

industry	industry_count
Business Products & Services	491
Advertising & Marketing	489
Software	461
Health	354
Construction	346
Consumer Products & Services	311
IT Management	276
Financial Services	239
Government Services	235
Real Estate	198

LEAST REPRESENTED

```
SELECT industry,
(COUNT(*)) AS industry_count
FROM inc_5000
GROUP BY industry
ORDER BY industry_count
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

industry	industry_count
Computer Hardware	32
Environmental Services	43
IT Services	43
Media	46
Travel & Hospitality	57
Security	66
Insurance	70
Education	70
Energy	76
Telecommunications	78

-- 2. SHOWING THE GEOGRAPHIC TRENDS USING REVENUE AMONG CITIES

```
SELECT city,
SUM(revenue) AS total_revenue
FROM Inc_5000
GROUP BY city
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city	total_revenue
Atlanta	4337500000
New York	4126100000
Chicago	3020700000
Houston	2701400000
San Diego	2275800000
Austin	2121700000
Dallas	2102500000
San Francisco	1812200000
Miami	1587600000
Boston	1559900000

-- 3. SHOWING AVERAGE REVENUE AMONG COMAPANIES BROKEN DOWN BY INDUSTRY

```
SELECT industry,
ROUND(AVG(revenue),0)  AS avg_revenue
FROM inc_5000
GROUP BY industry
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

industry	avg_revenue
Advertising & Marketing	18908384
Logistics & Transportation	54618132
Food & Beverage	39855906
Consumer Products & Services	24761093
Retail	36323602
Health	45540960
Telecommunications	29541026
Financial Services	40004184
Computer Hardware	31178125
Software	21573970

-- 4. SHOWING WHICH INDUSTIES SAW THE LARGEST AVERAGE GROWTH RATE

```
SELECT industry,
ROUND(AVG(`growth_%`), 2) AS avg_growth
FROM inc_5000
GROUP BY industry
ORDER BY avg_growth DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

industry	avg_growth
Logistics & Transportation	731.49
Consumer Products & Services	692.21
Food & Beverage	668.23
Computer Hardware	651.98
Energy	605.17
Retail	586.83
Health	548.75
Software	524.67
Telecommunications	521.14
Security	493.75

-- 5. SHOWING IF ANY COMAPANY INCREASED REVENUE WHILE REDUCING STAFF

```
WITH updated_inc_5000 AS (
SELECT name, founded,
`growth_%`,
ROUND((((workers - previous_workers) / previous_workers) * 100),2) AS pct_staff_turnover
FROM  inc_5000
)
SELECT
name, founded, `growth_%`, pct_staff_turnover
FROM updated_inc_5000
WHERE `growth_%` > 0
AND pct_staff_turnover < 0
ORDER BY founded DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	founded	growth_%	pct_staff_turnover
Alpha Premier	2014	126.375	-20.00
Hasko Trading	2014	67.2422	-52.17
Driftless Glen Distillery	2014	685.1916	-16.67
Adapture	2013	81.7726	-10.53
Synapse Business Systems	2013	204.0911	-100.00
TapClassifieds Auto	2013	74.9278	-20.83
Vennli	2013	109.6291	-42.86
360 Risk Partners Insurance Solutions	2012	365.3979	-44.44
Zentra	2012	235.0994	-30.00
RJR Partners	2012	105.1348	-13.33

-- 6. SHOWING WHICH COMPANIES THAT HAD THE LARGEST INCREASE IN STAFF/NEW HIRES

```
SELECT name,
(workers - previous_workers) AS staff_turnover
FROM inc_5000
ORDER BY staff_turnover DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

name	staff_turnover
Alorica	43000
Infiniti HR	37145
CircusTrix	13739
ScribeAmerica	11641
TaskUs	9198
Freddy's Frozen Custard	6494
Pathway Vet Alliance	4174
Four Foods Group	3500
The Revenue Optimization Companies	3239
Vaco	3022

-- 7. SHOWING INDUSTRIES AVERAGE STAFF TURNOVER

```
SELECT industry,
ROUND(AVG(workers - previous_workers),2) AS avg_staff_turnover
FROM inc_5000
GROUP BY industry
ORDER BY avg_staff_turnover DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

industry	avg_staff_turnover
Human Resources	403.10
Food & Beverage	251.30
Health	187.77
Business Products & Services	181.20
Insurance	117.79
Education	115.53
Security	94.26
Real Estate	91.59
Government Services	85.55
IT System Development	83.88

-- 8. SHOWING WHICH INDUSTRIES HAD THE LARGEST STAFF TURNOVER

```
SELECT industry,
sum(workers - previous_workers) AS staff_turnover
FROM inc_5000
GROUP BY industry
ORDER BY staff_turnover DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

industry	staff_turnover
Business Products & Services	88969
Health	66470
Human Resources	62481
Advertising & Marketing	32599
Software	32230
Food & Beverage	31915
Government Services	20104
Construction	19052
Financial Services	18145
Real Estate	18135

-- 9. CALCULATING THREE PERFORMANCE METRICS- TOTAL REVENUE, AVERAGE REVENUE, TOTAL NUMBER OF COMPANIES IN EACH INDUSTRIES

```
SELECT industry,
COUNT(DISTINCT name) as company_count,
SUM(revenue) AS total_revenue,
ROUND(AVG(revenue),2) AS avg_revenue
FROM inc_5000
GROUP BY industry
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

industry	company_count	total_revenue	avg_revenue
Health	354	16121500000	45540960.45
Business Products & Services	491	13072100000	26623421.59
Construction	346	12747900000	36843641.62
Software	461	9945600000	21573969.63
Logistics & Transportation	182	9940500000	54618131.87
Financial Services	239	9561000000	40004184.10
Advertising & Marketing	489	9246200000	18908384.46
Government Services	235	8616500000	36665957.45
Consumer Products & Services	311	7700700000	24761093.25
Real Estate	198	6744100000	34061111.11

-- 10. SHOWING TOP 10 INDUSTRIES WITH THE HIGHEST REVENUE AND 10 INDUSTRIES WITH THE LOWEST REVENUE

```
SELECT industry,
SUM(revenue) AS total_revenue
FROM inc_5000
GROUP BY industry
ORDER BY total_revenue DESC
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

industry	total_revenue
Health	16121500000
Business Products & Services	13072100000
Construction	12747900000
Software	9945600000
Logistics & Transportation	9940500000
Financial Services	9561000000
Advertising & Marketing	9246200000
Government Services	8616500000
Consumer Products & Services	7700700000
Real Estate	6744100000

10 INDUSTRIES WITH THE LOWEST REVENUE

```
SELECT industry,
SUM(revenue) AS sum_revenue
FROM inc_5000
GROUP BY industry
ORDER BY sum_revenue
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

industry	sum_revenue
Media	768600000
IT Services	822600000
Environmental Services	967800000
Computer Hardware	997700000
Education	1458500000
Travel & Hospitality	1524000000
Telecommunications	2304200000
Security	2491100000
IT System Development	2530600000
Engineering	2669300000